

Structural bioinformatic

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```
stats <- read.csv("C:/Users/chenr/Downloads/pdb_stats.csv")
stats
```

```
##           Molecular.Type X.ray    EM    NMR Integrative Multiple.methods
## 1           Protein (only) 178795 21825 12773           343           226
## 2 Protein/Oligosaccharide 10363  3564   34           8           11
## 3           Protein/NA    9106  6335  287           24           7
## 4      Nucleic acid (only)  3132   221 1566           3          15
## 5                Other    175    25   33           4           0
## 6 Oligosaccharide (only)    11     0    6           0           1
##   Neutron Other  Total
## 1     84     32 214078
## 2      1      0 13981
## 3      0      0 15759
## 4      3      1  4941
## 5      0      0   237
## 6      0      4    22
```

PDB statistics

```
stats$Total
```

```
## [1] 214078 13981 15759 4941 237 22
```

```
library(readr)
```

```
stats <- read_csv("C:/Users/chenr/Downloads/pdb_stats.csv")
```

```
## Rows: 6 Columns: 9
```

```
## -- Column specification -----
```

```
## Delimiter: ","
```

```
## chr (1): Molecular Type
```

```
## dbl (8): X-ray, EM, NMR, Integrative, Multiple methods, Neutron, Other, Total
```

```
##
```

```
## i Use `spec()` to retrieve the full column specification for this data.
```

```
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
n.total <- sum(stats$Total)
```

```
n.total
```

```
## [1] 249018
```

Q1: What percentage of structures in the PDB are solved by X-Ray and Electron Microscopy.

```
n.xray <- sum(stats$`X-ray`)
```

```
round(n.xray / n.total * 100)
```

```
## [1] 81
```

```
n.xray <- sum(stats$`X-ray`)  
percent.xray <- n.xray / n.total * 100  
percent.xray
```

```
## [1] 80.95077
```

80.95% of structures in the PDB are solved by X-Ray and Electron Microscopy. There are 81.43 percent Xray structures in the PDB

Q2: What proportion of structures in the PDB are protein?

```
round(stats$Total[1]/n.total*100, 2)
```

```
## [1] 85.97
```

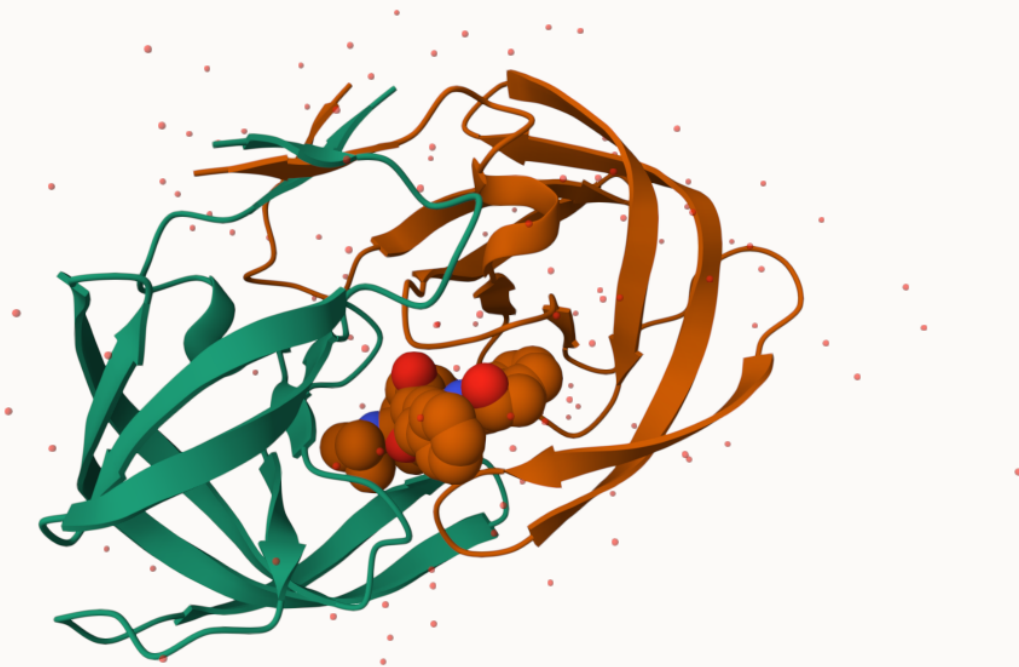
85.97 of structures in the PDB are protein

Q3: Type HIV in the PDB website search box on the home page and determine how many HIV-1 protease structures are in the current PDB?

Current HIV-1 protease structure in the PDB at May 10th, 2026: 1141

The PDB format

```
library(png)  
library(grid)  
  
img1 <- readPNG("C:/Users/chenr/Downloads/1HSG.png")  
grid.newpage()  
grid.raster(img1)
```



HIV with Mol Reviewer

Q4: Water molecules normally have 3 atoms. Why do we see just one atom per water molecule in this structure?

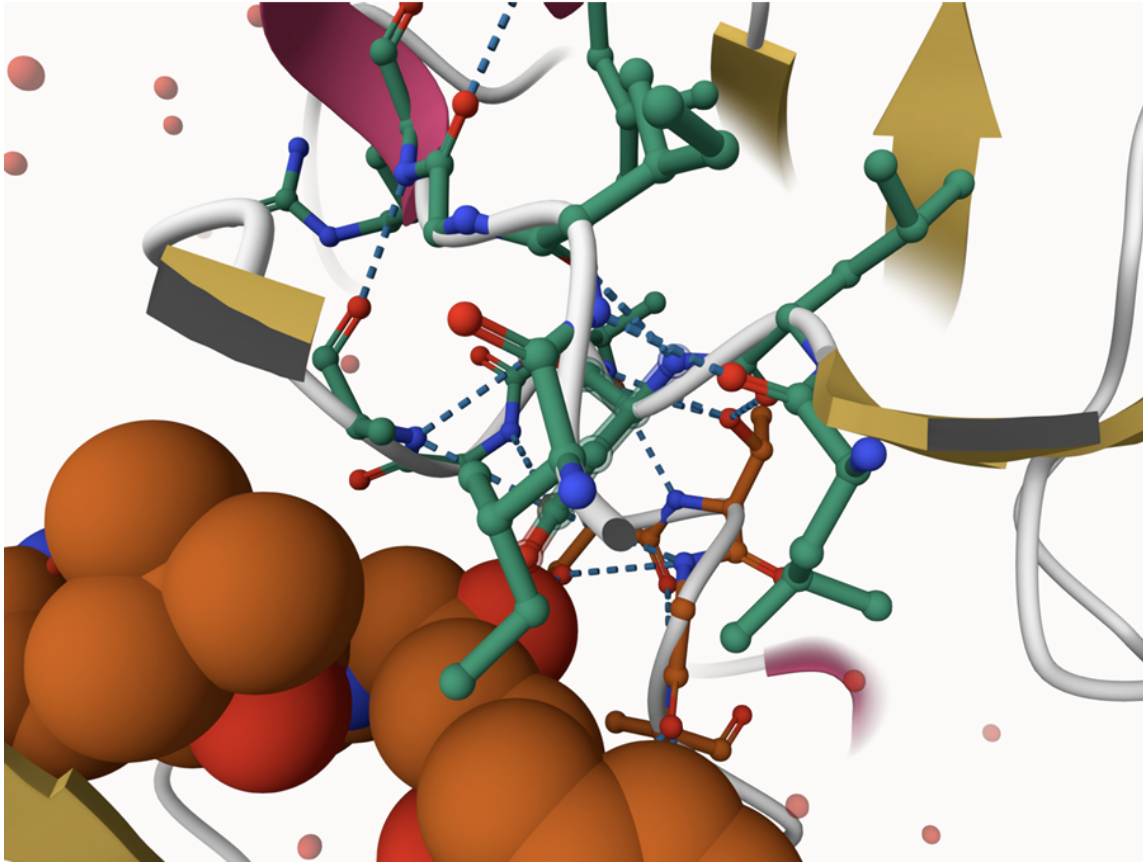
In X-ray crystal structures like HIV-1 Protease (1HSG), only water oxygen atoms are recorded because hydrogen atoms are usually not detected by X-ray crystallography.

Q5: There is a critical “conserved” water molecule in the binding site. Can you identify this water molecule? What residue number does this water molecule have?

In HIV-1 Protease (1HSG), HOH308 is likely the conserved water molecule because it is located near both the ligand and the catalytic Asp25 residues, a typical feature of the conserved active-site water in HIV-1 protease.

Q6: Generate and save a figure clearly showing the two distinct chains of HIV protease along with the ligand. You might also consider showing the catalytic residues ASP 25 in each chain and the critical water (we recommend “Ball & Stick” for these side-chains). Add this figure to your Quarto document.

```
img2 <- readPNG("C:/Users/chenr/Downloads/1hsg view of ligand.png")
grid.newpage()
grid.raster(img2)
```



Ligand Binding Site Showing Catalytic Asp25 Residues and Conserved Water Molecule

Indinavir and other large ligands enter the HIV-1 Protease (1HSG) binding site through flexible flap regions that open to expose the catalytic site and close to stabilize the ligand during binding.

Reading PDB file data into R

```
library(bio3d)
hiv <- read.pdb("1hsg")
```

```
## Note: Accessing on-line PDB file
```

```
hiv
```

```
##
```

```
## Call: read.pdb(file = "1hsg")
```

```
##
```

```
## Total Models#: 1
```

```
## Total Atoms#: 1686, XYZs#: 5058 Chains#: 2 (values: A B)
```

```
##
```

```
## Protein Atoms#: 1514 (residues/Calpha atoms#: 198)
```

```
## Nucleic acid Atoms#: 0 (residues/phosphate atoms#: 0)
```

```
##
```

```
## Non-protein/nucleic Atoms#: 172 (residues: 128)
```

```
## Non-protein/nucleic resid values: [ HOH (127), MK1 (1) ]
```

```
##
```

```
## Protein sequence:
```

```
## PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWPKMIGGIGGFIKVRQYD
```

```
##      QILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFPQITLWQRPLVTIKIGGQLKE
##      ALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYDQILIEICGHKAIGTVLVGPTP
##      VNIIGRNLLTQIGCTLNF
##
## + attr: atom, xyz, seqres, helix, sheet,
##      calpha, remark, call
```

Q7: How many amino acid residues are there in this pdb object?

```
sum(hiv$calpha)
```

```
## [1] 198
```

198 residues

Q8: Name one of the two non-protein residues?

```
unique(hiv$atom$resid[hiv$atom$type== "HETATM"])
```

```
## [1] "MK1" "HOH"
```

MK1(indinavir) or water

Q9: How many protein chains are in this structure?

```
unique(hiv$atom$chain[hiv$atom$type== "ATOM"])
```

```
## [1] "A" "B"
```

```
length(unique(hiv$atom$chain[hiv$atom$type== "ATOM"]))
```

```
## [1] 2
```

2chains(A and B)

```
attributes(hiv)
```

```
## $names
## [1] "atom" "xyz" "seqres" "helix" "sheet" "calpha" "remark" "call"
##
## $class
## [1] "pdb" "sse"
```

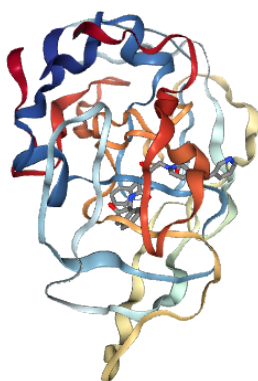
```
head(hiv$atom)
```

```
##   type eleno elety alt resid chain resno insert      x      y      z o      b
## 1 ATOM     1     N <NA>  PRO     A      1 <NA> 29.361 39.686 5.862 1 38.10
## 2 ATOM     2     CA <NA>  PRO     A      1 <NA> 30.307 38.663 5.319 1 40.62
## 3 ATOM     3     C <NA>  PRO     A      1 <NA> 29.760 38.071 4.022 1 42.64
## 4 ATOM     4     O <NA>  PRO     A      1 <NA> 28.600 38.302 3.676 1 43.40
## 5 ATOM     5     CB <NA>  PRO     A      1 <NA> 30.508 37.541 6.342 1 37.87
## 6 ATOM     6     CG <NA>  PRO     A      1 <NA> 29.296 37.591 7.162 1 38.40
##   segid elesy charge
## 1 <NA>     N  <NA>
## 2 <NA>     C  <NA>
## 3 <NA>     C  <NA>
## 4 <NA>     O  <NA>
## 5 <NA>     C  <NA>
## 6 <NA>     C  <NA>
```

```
library(bio3dview)
```

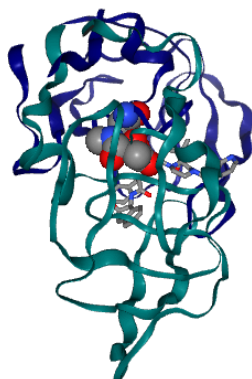
```
library(NGLViewerR)
```

```
view.pdb(hiv) |>  
  setSpin()
```



```
# Select the important ASP 25 residue  
sele <- atom.select(hiv, resno=25)
```

```
# and highlight them in spacefill representation  
view.pdb(hiv, cols=c("navy","teal"),  
         highlight = sele,  
         highlight.style = "spacefill") |>  
setRock()
```

Predicting functional motions of a single structure Let's read a new PDB structure of Adenylate Kinase and perform Normal mode analysis.

```

adk <- read.pdb("6s36")

## Note: Accessing on-line PDB file
## PDB has ALT records, taking A only, rm.alt=TRUE

adk

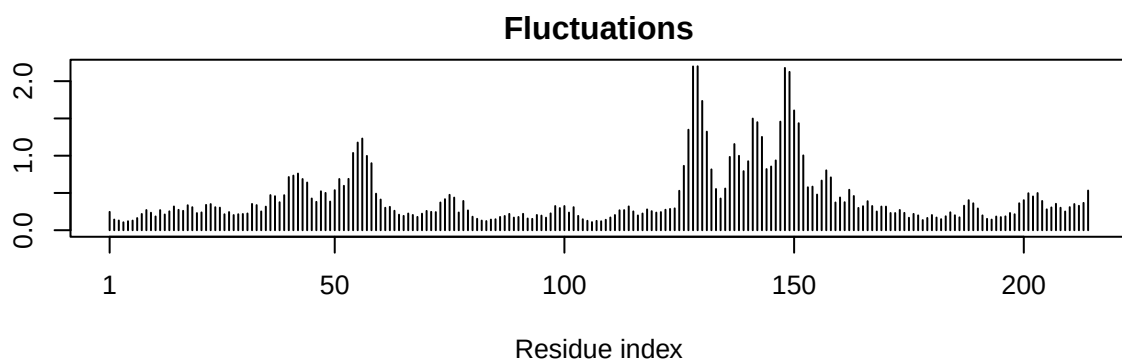
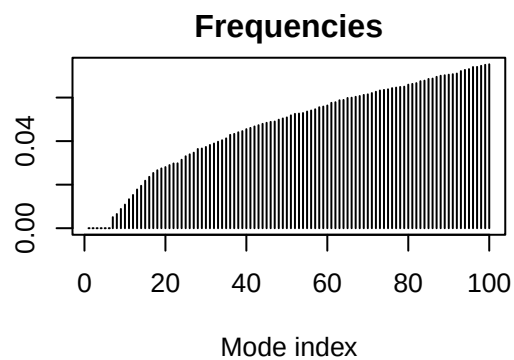
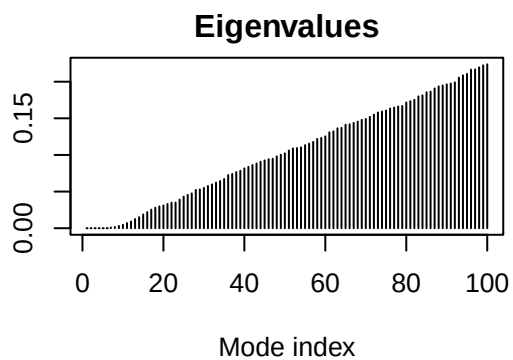
##
## Call: read.pdb(file = "6s36")
##
## Total Models#: 1
## Total Atoms#: 1898, XYZs#: 5694 Chains#: 1 (values: A)
##
## Protein Atoms#: 1654 (residues/Calpha atoms#: 214)
## Nucleic acid Atoms#: 0 (residues/phosphate atoms#: 0)
##
## Non-protein/nucleic Atoms#: 244 (residues: 244)
## Non-protein/nucleic resid values: [ CL (3), HOH (238), MG (2), NA (1) ]
##
## Protein sequence:
## MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAAVKSGSELGKQAKDIMDAGKLVT
## DELVIALVKERIAQEDCRNGFLLDGFPRTIPQADAMKEAGINVDYVLEFDVPDELIVDKI
## VGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKDDQEETVRKRLVEYHQMTAPLIG
## YYSKEAEAGNTKYAKVDGTPVAEVRADLEKILG
##
## + attr: atom, xyz, seqres, helix, sheet,
## calpha, remark, call

m <- nma(adk)

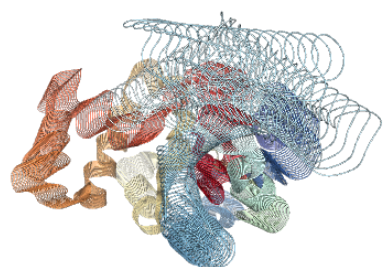
## Building Hessian... Done in 0.05 seconds.
## Diagonalizing Hessian... Done in 0.33 seconds.

plot(m)

```



```
mktrj(m, file="adk_m7.pdb")  
view.nma(m)
```



Comparative structure analysis of Adenylate Kinase

```
# Install packages in the R console NOT your Rmd/Quarto file
#install.packages("bio3d")
#install.packages("NGLVieweR")
#install.packages("pak")
#pak::pak("bioboot/bio3dview")
#install.packages("BiocManager")
#BiocManager::install("msa")
```

Q10. Which of the packages above is found only on BioConductor and not CRAN?

msa. It's a Bioconductor package (installed via `BiocManager::install("msa")`), not on CRAN

Q11. Which of the above packages is not found on BioConductor or CRAN?

bio3dview. It's GitHub-only (`pak::pak("bioboot/bio3dview")`), not on CRAN or Bioconductor

Q12. True or False? Functions from the pak package can be used to install packages from GitHub and BitBucket?

True. pak can install from GitHub and Bitbucket using remote specs (e.g., "user/repo" for GitHub, `bitbucket::user/repo`).

Search and retrieve ADK structures

```
aa <- get.seq("1ake_A")
```

```
## Warning in get.seq("1ake_A"): Removing existing file: seqs.fasta
```

```
## Fetching... Please wait. Done.
```

```
aa
```

```
##           1           .           .           .           .           .           60
## pdb|1AKE|A MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMRLRAAVKSGSELGKQAKDIMDAGKLV
##           1           .           .           .           .           .           60
##
##           61           .           .           .           .           .           120
## pdb|1AKE|A DELVIALVKERIAQEDCRNGFLLDGFPRTIPQADAMKEAGINVDYVLEFDVPDELIVDRI
##           61           .           .           .           .           .           120
##
##           121          .           .           .           .           .           180
## pdb|1AKE|A VGRRVHAPSGRVYHVKFNPPKVEGKDDVTGEELTTRKDDQEETVRKRLVEYHQM
##           121          .           .           .           .           .           180
##
##           181          .           .           .           .           .           214
## pdb|1AKE|A YYSKEAEAGNTKYAKVDGTPVAEVRADLEKILG
##           181          .           .           .           .           .           214
##
## Call:
##   read.fasta(file = outfile)
##
## Class:
##   fasta
##
## Alignment dimensions:
##   1 sequence rows; 214 position columns (214 non-gap, 0 gap)
```

##

+ attr: id, ali, call

Q13. How many amino acids are in this sequence, i.e. how long is this sequence?

214 amino acids.